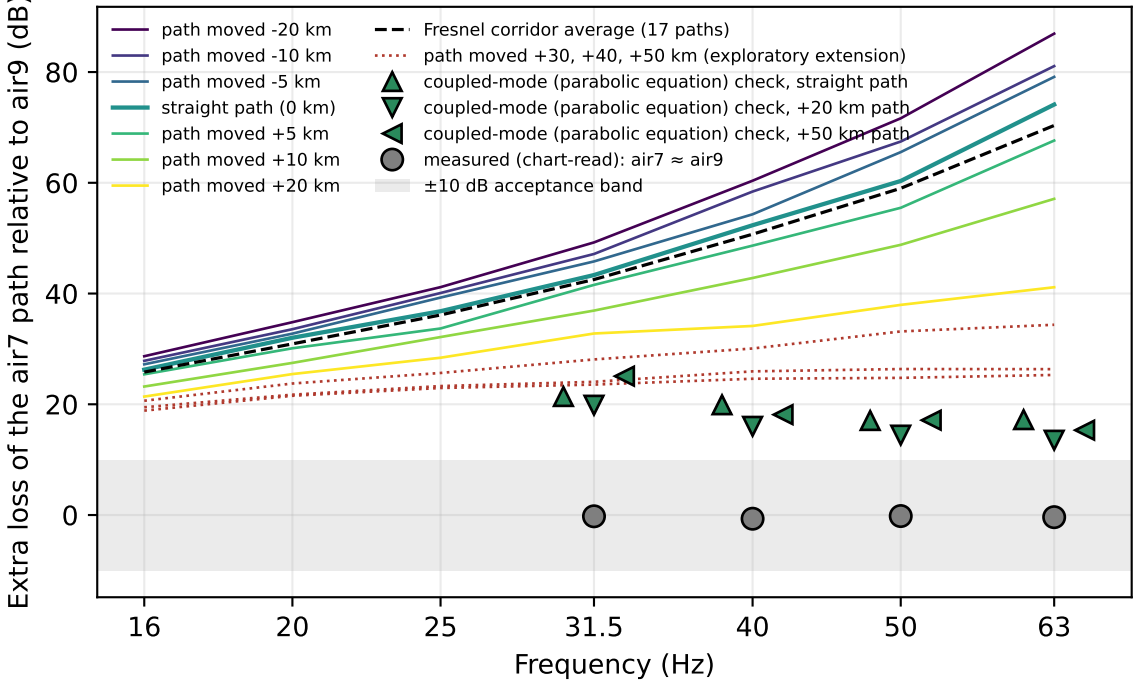
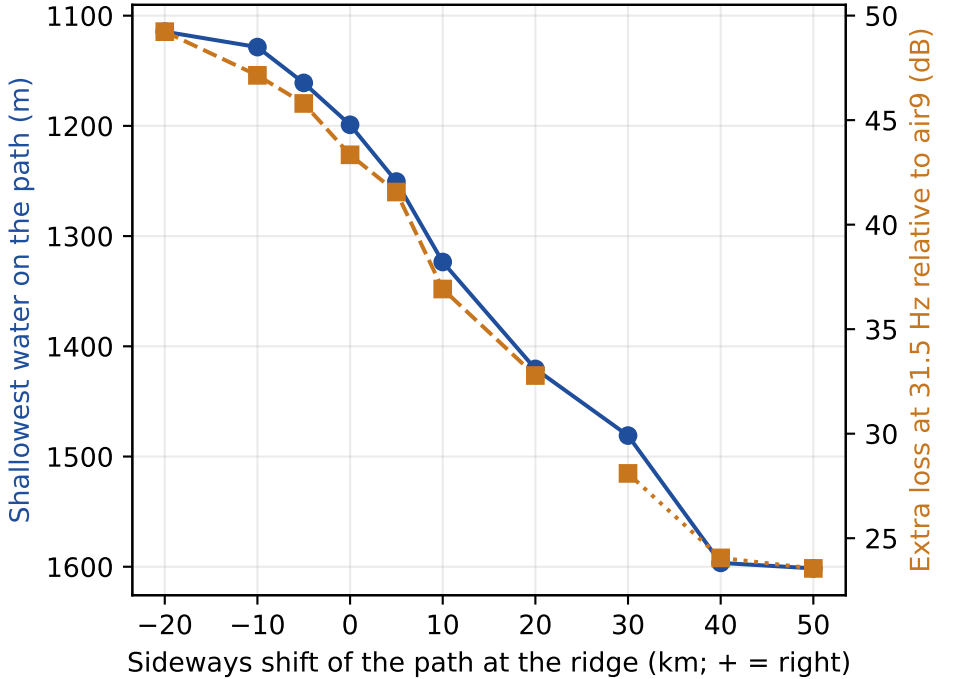
Airgun site air7 (October 2001) to Cape Leeuwin: can a path beside the ridge explain that air7 was heard as well as air9?

a Every mode-by-mode path stays 33 dB or more above the measurement; the coupled-mode check brings the straight path down to about 20 dB



b Shifting the path right deepens the sill and lowers the loss, but about 24 dB of excess remains once the ridge is cleared



What the chart shows: the shots at site air7 were heard at Cape Leeuwin about as well as the shots at site air9, yet our model says the air7 path loses 43–74 dB more because it crosses a ridge 1,190 m deep. (a) Moving the path sideways by up to 20 km, or averaging over a corridor of paths, does not close the gap; the best path is still 33–41 dB too lossy. A second model that lets sound pass over the ridge more realistically (green) brings the straight path to 17–22 dB too lossy. (b) Moving 40–50 km to the right clears the ridge, but about 24 dB of extra loss remains. Verdict under the rule fixed in advance: NOT REPRODUCED. The measurement itself may be biased: only shots heard above the noise were plotted. Caveats C1 and C2 apply.

Technical: KRAKEN adiabatic incoherent modes, hard bottom, Francois-Garrison, z_s 10 m, receiver 1,055 m; source = shot-weighted mean of the air7 segments (22.913 S 87.018 E); sections GEBCO_2026 3×3 median at 0.5 km, WOA23 95A4 October (audit path builder). Offsets d(s) = d_max sin(πs/R). N×2D = nx2d.py Fresnel corridor (17 members, exp(−u²) weights, all members open). Reference: air9 → H01W on the same builder. Measured: Fig. 23 CL panel air7 and air9 band means (DIGITISED, airgun_sites_confirm). Medians over 31.5–63 Hz of (model − observed) difference: straight 56.8, corridor 55.3, least-loss ±20 km 36.5 dB. Pre-registration e949144 (amendment 2 adds +30/+40/+50 km; amendment 3 the RAM set; both exploratory). RAM = pyram split-step Padé, dr 2λ, dz λ/20, 9 frequencies per band, no absorption in either model for that comparison; RAM medians over 31.5–63 Hz: straight +19.1, +20 km +15.7, +50 km +18.1 dB. Gap G22. No MH370 posterior enters; C1/C2 stay on every MH370 number.